

Chapter 1C: Acute Coronary Syndromes

Purpose of this chapter: This is original Medvale teaching text. The uploaded cardiology pages and Anki-derived signals are used as coverage and high-yield clue prompts only, not as source text to copy.

Why this matters clinically

Acute coronary syndrome is the emergency version of coronary artery disease. Stable angina is usually a predictable supply-demand mismatch from fixed coronary narrowing. ACS is different: it usually reflects acute plaque disruption with platelet activation, thrombus formation, and sudden reduction in coronary blood flow.

The first clinical task is to route the patient into the correct emergency pathway: unstable angina, NSTEMI, or STEMI. These categories are not just vocabulary. They determine monitoring, antiplatelet therapy, anticoagulation, catheterization timing, and whether immediate reperfusion is needed.

ACS category	ECG	Troponin	Core meaning
Unstable angina	No persistent ST elevation	Negative	Ischemia without detectable myocardial necrosis
NSTEMI	No persistent ST elevation	Positive	Myocardial necrosis without STEMI ECG pattern
STEMI	ST elevation or equivalent	Often positive, but may be early negative	Acute coronary occlusion until proven otherwise

Medvale rule. STEMI is an ECG emergency. NSTEMI is a biomarker-positive ACS. Unstable angina is an ACS pattern without detectable necrosis.

Core illness scripts

Unstable Angina

Category	Illness script
Mechanism	Plaque disruption or worsening coronary obstruction causing ischemia without detectable necrosis
Symptoms	New, worsening, rest, prolonged, or lower-threshold angina
ECG	May show ST depression, T-wave inversion, transient changes, or be nondiagnostic
Troponin	Negative
Site of care	Hospital/monitored setting
Key danger	Can progress to NSTEMI/STEMI

NSTEMI

Category	Illness script
Mechanism	Acute coronary thrombosis with incomplete occlusion or distal embolization
Symptoms	Ischemic chest pain, dyspnea, diaphoresis, nausea; atypical symptoms possible
ECG	ST depression, T-wave inversion, transient ST elevation, or nondiagnostic
Troponin	Positive with rise/fall pattern
Site of care	Hospital; monitored setting
Key decision	Early invasive versus ischemia-guided strategy

STEMI

Category	Illness script
Mechanism	Acute coronary occlusion causing transmural ischemia
Symptoms	Persistent severe ischemic chest discomfort, diaphoresis, dyspnea, nausea
ECG	ST elevation in contiguous leads or STEMI equivalent
Troponin	May be negative early; later positive
Site of care	Emergency reperfusion pathway
Key decision	Immediate PCI if available; fibrinolysis if PCI delay and no contraindications

Numbers to Know: ACS

Topic	High-yield number
Chest pain duration concerning for MI	Persistent pain >20-30 minutes
Aspirin loading dose in many classic ACS pathways	162-325 mg chewed, unless contraindicated
Aspirin dose shown in uploaded source	325 mg
High-intensity atorvastatin dose in uploaded source	40-80 mg early after ACS
Unfractionated heparin/LMWH duration in older ACS pathways	At least 48 hours or until PCI strategy complete
Early invasive strategy for many high-risk NSTEMI-ACS patients	Often within 24 hours when high-risk features are present
Older teaching from source for many UA/NSTEMI patients	Cath/revascularization within 1-2 days
Troponin I/T begins rising	About 3-5 hours after injury
Troponin peak	Around 24-48 hours
Troponin remains elevated	About 5-14 days
CK-MB begins rising	About 4-8 hours
CK-MB peak	Around 24 hours
CK-MB returns to normal	About 48-72 hours
Bare-metal stent minimum DAPT teaching	At least 30 days
Drug-eluting stent minimum DAPT teaching	Often at least 12 months in classic ACS teaching
Post-MI free wall rupture timing	Usually 1-14 days, commonly 3-7 days
Dressler syndrome timing	Weeks to months after MI

These numeric anchors are included because they commonly drive exam questions and management decisions: medication dose choices, biomarker interpretation, timing of catheterization, and post-MI complication windows.

Pathophysiology

ACS begins with an unstable coronary plaque. The fibrous cap disrupts, exposing thrombogenic material to the bloodstream. Platelets adhere, activate, and aggregate. The coagulation cascade generates fibrin, and a thrombus forms.

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Atherosclerotic plaque
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Plaque rupture or erosion
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Platelet activation + thrombin generation
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Coronary thrombus
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Partial occlusion -> UA/NSTEMI
Complete occlusion -> STEMI

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The difference between unstable angina and NSTEMI is not the severity of pain. It is whether myocardial cell death is detectable. NSTEMI has troponin elevation; unstable angina does not.

Presentation patterns

Symptom	Description
Chest pressure	Substernal heaviness, squeezing, crushing, tightness
Radiation	Left arm, both arms, neck, jaw, back, epigastrium
Autonomic symptoms	Diaphoresis, nausea, vomiting
Dyspnea	May be the main symptom
Weakness/fatigue	Common in older adults
Syncope	Suggests arrhythmia, shock, severe ischemia, PE, or dissection
Sense of doom	High-yield exam phrase, not diagnostic alone

Atypical ACS is common in older adults, diabetic patients, women, postoperative patients, and patients with chronic kidney disease. Symptoms may be dyspnea, nausea, epigastric discomfort, fatigue, or syncope rather than obvious crushing chest pain.

Medvale rule. Do not require classic chest pain to diagnose ACS in a high-risk patient.

Initial ED approach

Step	Action	Why
1	ABCs, vitals, IV access, cardiac monitor	Detect instability and arrhythmias
2	12-lead ECG	Identify STEMI or ischemia
3	Repeat ECGs if symptoms continue	Initial ECG can be nondiagnostic
4	Troponin now and serially	Detect myocardial injury
5	Aspirin if no contraindication	Early antiplatelet benefit
6	Anticoagulation if ACS likely	Prevent thrombus propagation
7	Risk stratify	Decide invasive timing
8	Reperfusion if STEMI	Time-sensitive myocardial salvage

ECG reasoning in ACS

STEMI localization

Territory	Leads	Common artery
Inferior	II, III, aVF	RCA most often; sometimes LCX
Septal/anterior	V1-V4	LAD
Lateral	I, aVL, V5-V6	LCX or diagonal branch

Posterior	ST depression V1-V3 with tall R waves; confirm V7-V9	RCA or LCX
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NSTEMI/unstable angina ECG patterns

ECG finding	Meaning
ST depression	Subendocardial ischemia
T-wave inversion	Ischemia; sensitive but not specific
Transient ST elevation	Dynamic coronary occlusion/reperfusion possible
Normal ECG	Does not exclude ACS
New arrhythmia	May reflect ischemia or complication

Troponin and CK-MB

Troponin

Troponin feature	High-yield point
Starts rising	3-5 hours after injury
Peaks	24-48 hours
Remains elevated	5-14 days
Best use	Diagnosis of myocardial injury
Limitation	Elevation is not specific for type 1 plaque-rupture MI

CK-MB

CK-MB feature	High-yield point
Starts rising	4-8 hours
Peaks	Around 24 hours
Returns to baseline	48-72 hours
Best use	Detecting reinfarction soon after initial MI

Exam clue. If a patient has recurrent chest pain shortly after an MI, troponin may still be elevated from the first infarct. CK-MB can be useful because it returns to normal sooner.

ACS classification table

Feature	Unstable angina	NSTEMI	STEMI
Rest pain possible	Yes	Yes	Yes
Troponin positive	No	Yes	Often yes, may be early negative
Persistent ST elevation	No	No	Yes
Myocardial necrosis	No detectable necrosis	Yes	Yes
Immediate reperfusion pathway	No	Usually no, unless unstable/high-risk anatomy	Yes
Hospital admission	Yes	Yes	Yes
Antiplatelet therapy	Yes, if no contraindication	Yes	Yes
Anticoagulation	Usually yes	Yes	Yes
Cath timing	Based on risk	Based on risk	Immediate PCI if available

Initial ACS medication bundle

Antiplatelet therapy

Medication	Role	High-yield notes
Aspirin	Blocks thromboxane A2 platelet activation	Chew early unless allergy or active major bleeding
P2Y12 inhibitor	Blocks ADP-mediated platelet activation	Clopidogrel, ticagrelor, prasugrel
GP IIb/IIIa inhibitor	Blocks final common platelet aggregation pathway	Selected high-risk PCI/thrombus situations; less routine than older teaching

Anticoagulation

Medication	Role
Unfractionated heparin	Anticoagulation during ACS/PCI pathways
Enoxaparin	LMWH option; older source notes superiority over UFH in some UA/NSTEMI contexts
Bivalirudin	Alternative in selected PCI settings

Anticoagulation prevents extension of thrombus. It does not dissolve the plaque; it helps prevent clot propagation while antiplatelet and invasive strategies address the occlusive process.

Anti-ischemic therapy

Medication	Mechanism	Caution
Nitroglycerin	Venodilation lowers preload; coronary dilation may help symptoms	Avoid hypotension, RV infarct, recent PDE-5 inhibitor use
Beta-blocker	Lowers HR, BP, contractility, oxygen demand	Avoid acute decompensated HF, shock, severe bradycardia/high-grade AV block
Morphine	Analgesia, venodilation	May mask worsening symptoms and may affect antiplatelet absorption
Oxygen	Helps if hypoxemic	Not routinely needed if normal oxygen saturation

Statins

High-intensity statin therapy is started early unless contraindicated. The uploaded cardiology source highlights atorvastatin 40-80 mg started early after ACS.

STEMI pathway

STEMI is a reperfusion diagnosis. The ECG and clinical context trigger action; do not wait for troponin if the presentation is convincing.

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STEMI or STEMI equivalent
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Can PCI be performed rapidly?
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Yes -> Primary PCI
No -> Check fibrinolysis eligibility
|
No contraindications + appropriate time window
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Fibrinolysis, then transfer for angiography/rescue PCI if needed
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Option	Best use
Primary PCI	Preferred when available rapidly
Fibrinolysis	STEMI when PCI delay is too long and no contraindications
CABG	Selected anatomy, failed PCI, mechanical complications, or complex disease

NSTEMI / unstable angina pathway

NSTEMI and unstable angina require hospitalization, monitoring, serial ECG/troponins, medical therapy, and risk stratification.

High-risk features favoring early invasive strategy

High-risk feature	Why it matters
Hemodynamic instability	Possible large ischemic territory or shock
Recurrent or refractory chest pain	Ongoing ischemia
Dynamic ST changes	Active ischemia
Rising troponin	Myocardial injury
Ventricular arrhythmias	Electrical instability
New heart failure	Large territory ischemia or mechanical dysfunction
New mitral regurgitation	Possible papillary muscle ischemia/rupture
High-risk score	Higher short-term event risk

TIMI-style risk thinking

Risk scores help decide how aggressive the early strategy should be. A higher-risk NSTEMI-ACS patient is more likely to need early invasive evaluation. A lower-risk patient who stabilizes with medical therapy may undergo a more selective strategy.

Risk concept	Pushes toward
Recurrent ischemia	Early invasive evaluation
Positive troponin	Early invasive evaluation
ST depression	Higher risk
Diabetes/CKD/prior CAD	Higher baseline risk
Hemodynamic instability	Immediate invasive/ICU-level care
Low-risk features, negative serial troponins, stable ECG	Observation or selective testing

Important contraindication thinking

Nitrates

Avoid/caution situation	Why
Hypotension	Can worsen shock
Right ventricular infarction	Patient may be preload dependent
Recent PDE-5 inhibitor use	Severe hypotension
Severe aortic stenosis	Preload reduction may cause collapse

Beta-blockers

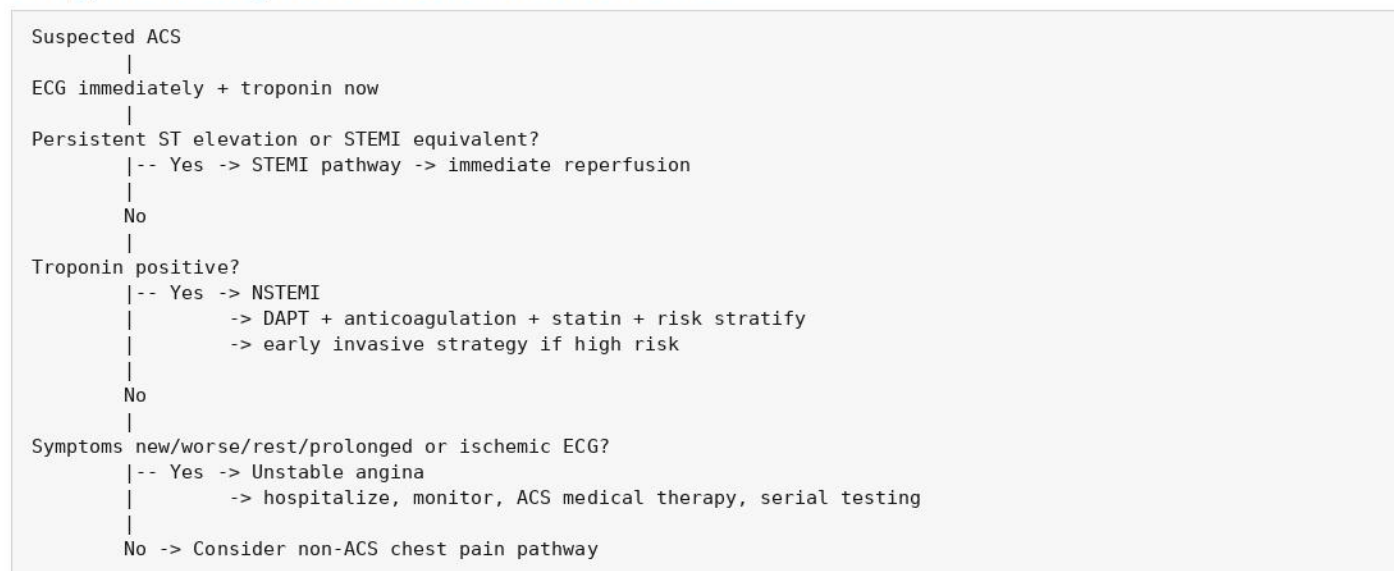
Avoid/caution situation	Why
Cardiogenic shock	Negative inotropy worsens perfusion
Acute decompensated HF	Can worsen pump failure
Severe bradycardia	Further slows HR
High-grade AV block	Worsens conduction
Severe reactive airway disease	Bronchospasm risk, especially nonselective agents

Fibrinolysis

Absolute contraindications classically include prior intracranial hemorrhage, known structural cerebral vascular lesion, known malignant intracranial neoplasm, recent ischemic stroke, suspected aortic dissection, active bleeding, significant closed-head/facial trauma, and severe uncontrolled hypertension depending on context.

Medvale rule. Never give fibrinolysis or routine anticoagulation for ACS until dissection is considered in a patient with tearing pain, pulse deficits, neurologic signs, or mediastinal widening.

Original diagram: ACS decision tree



Original table: ACS medication map

Goal	Medication	Number anchor
Platelet inhibition	Aspirin	Classic loading 162-325 mg; source lists 325 mg
Additional platelet inhibition	P2Y12 inhibitor	Clopidogrel, ticagrelor, prasugrel
Thrombus control	Heparin/LMWH	Source notes at least 48 hr in older UA/NSTEMI approach
Plaque stabilization	High-intensity statin	Atorvastatin 40-80 mg in source
Reduce demand	Beta-blocker	Use if no shock/HF/bradycardia/AV block
Pain/ischemia relief	Nitroglycerin	Avoid RV MI/hypotension/PDE-5 inhibitors
Hypoxemia correction	Oxygen	Use if hypoxic
Severe pain	Morphine	Controversial; use selectively

Common exam traps

Waiting for troponin in STEMI

A patient with classic symptoms and STEMI ECG changes needs reperfusion planning immediately. Troponin may be negative early.

Calling unstable angina stable because troponin is negative

Unstable angina has negative troponins. The diagnosis comes from the pain pattern and ischemic risk.

Giving nitrates in RV infarction

Inferior MI plus hypotension, clear lungs, and elevated JVP suggests RV infarction. These patients are preload dependent.

Forgetting CK-MB for reinfarction

Troponin stays elevated for days. CK-MB returns to normal faster and can help identify reinfarction shortly after an MI.

Missing posterior MI

ST depression in V1-V3 with tall R waves may be posterior MI, not just NSTEMI. Get posterior leads.

Anticoagulating a dissection

Chest pain radiating to the back with pulse deficits, neurologic symptoms, or severe hypertension should trigger dissection evaluation before routine anticoagulation.

Practice questions

Question 1

A 64-year-old man has crushing substernal chest pain for 45 minutes. ECG shows ST elevation in II, III, and aVF. Troponin is pending.

Answer: Activate the STEMI reperfusion pathway.

Teaching point: STEMI is treated based on ECG and clinical presentation; do not wait for troponin.

Question 2

A 70-year-old woman has rest chest pressure and diaphoresis. ECG shows ST depression in V4-V6. Troponin is elevated.

Answer: NSTEMI.

Teaching point: Ischemic symptoms plus positive troponin without STEMI ECG pattern equals NSTEMI.

Question 3

A 59-year-old man has new rest angina. ECG shows T-wave inversions. Serial troponins are negative.

Answer: Unstable angina.

Teaching point: Troponin-negative ACS exists. The unstable pain pattern matters.

Question 4

A patient has recurrent chest pain 2 days after an MI. Troponin remains elevated.

Answer: CK-MB is the best biomarker for reinfarction.

Teaching point: CK-MB returns to baseline in about 48-72 hours, while troponin may stay elevated 5-14 days.

Question 5

A patient with inferior STEMI is hypotensive with clear lungs and elevated JVP.

Answer: Avoid nitrates and aggressive diuretics.

Teaching point: This suggests right ventricular infarction with preload dependence.

Chapter summary

Acute coronary syndrome includes unstable angina, NSTEMI, and STEMI. Unstable angina has ischemic symptoms without troponin elevation; NSTEMI has troponin elevation without persistent STEMI ECG findings; STEMI is an ECG-defined reperfusion emergency. Initial management centers on ECG, serial troponins, aspirin, P2Y12 therapy, anticoagulation, statins, anti-ischemic therapy, and risk-based invasive evaluation.

Numeric anchors matter: troponin rises in 3-5 hours, peaks at 24-48 hours, and remains elevated 5-14 days; CK-MB rises in 4-8 hours, peaks around 24 hours, and normalizes in 48-72 hours. STEMI requires rapid reperfusion, while NSTEMI/unstable angina require monitoring, ACS medical therapy, and early invasive evaluation when high-risk features are present.

Source notes

Coverage checklist derived from the user-provided cardiology excerpts and Anki cardiology topic signals. The uploaded ACS/MI cardiology pages emphasize unstable angina, ACS treatment, variant angina, MI ECG localization, troponin and CK-MB timing, post-MI therapies, reperfusion, and post-MI complications. The prose and tables in this PDF are original Medvale writing.